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### Crane, and control method of crane

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#### Abstract

Provided is a crane and a control method capable of preventing an irregular winding from being caused, without significant deterioration in work efficiency. The crane includes an allowable deceleration rate derivation unit and a winch control unit. The allowable deceleration rate derivation unit derives an allowable deceleration rate representing an allowable value of the deceleration rate of the winding of a suspension rope, from a measured load that is a suspension load measured by a load measurement device. The allowable deceleration rate derivation unit derives the allowable deceleration rate that is decreased with a decrease in the measured load. The winch control unit makes the winding of the suspension rope by a winch device decelerated at a deceleration rate limited within a range equal to or less than the allowable deceleration rate.

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| <b>Inventors:</b>            | <b>Sakurai; Hitoshi (Hyogo, JP), Sasai; Shintaro (Hyogo, JP)</b> |
| <b>Applicant:</b>            | <b>KOBELCO CONSTRUCTION MACHINERY CO., LTD.</b> (Hiroshima, JP)  |
| <b>Family ID:</b>            | <b>1000008780535</b>                                             |
| <b>Assignee:</b>             | <b>KOBELCO CONSTRUCTION MACHINERY CO., LTD.</b> (Hiroshima, JP)  |
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*Primary Examiner:* Momper; Anna M

*Assistant Examiner:* Campos, Jr.; Juan J

*Attorney, Agent or Firm:* Oblon, McClelland, Maier & Neustadt, L.L.P.

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**Background/Summary**

## TECHNICAL FIELD

(1) The present invention relates to a crane capable of winding a suspension rope and a method for controlling the crane.

## BACKGROUND ART

(2) A typical crane includes a boom and a winch device. From the boom is suspended a suspended load through a suspension rope. The winch device hoists and lowers the suspended load by winding or unwinding the suspension rope. The winch device includes a winch drum around which the suspension rope is wound, and a motor for rotating the winch drum in a winding direction and an unwinding direction.

(3) There may occur an irregular winding in the crane. The irregular winding is a state where the winding of the suspension rope around the winch drum is disordered. The irregular winding may cause, for example, a temporary sudden drop of the suspended load.

(4) Patent Literature 1 discloses a crane including a control device for controlling a winch. In the crane, a hydraulic cylinder applies a constant load to the suspension rope through a link. The control device stops the winch when the angle of the link exceeds a predetermined angle during the unwinding of the suspension rope, thereby preventing the suspension rope from being excessively delivered upon the landing of the suspended load.

(5) When the load is light, however, the rapid deceleration of the hoisting of the suspension rope by the winch device may involve temporary looseness of the suspension rope to cause the irregular winding.

(6) On the other hand, slow deceleration of the winding of the suspension rope for preventing the irregular winding deteriorates the efficiency in the work of carrying the suspended load by the crane.

## CITATION LIST

### Patent Literature

(7) Patent Literature 1: Japanese Unexamined Patent Publication No. 2000-313592

## SUMMARY OF INVENTION

(8) It is an object of the present invention to provide a crane and crane control method that are capable of preventing an irregular winding from being caused in a winch device by rapid deceleration of the winding of a suspension rope without a significant deterioration in work efficiency.

(9) Provided is a crane including a boom, a winch device, a winch control unit, a load measurement device, and an allowable deceleration rate derivation unit. The boom supports a suspension rope suspended from the boom. The winch device is configured to perform winding and unwinding of the suspension rope. The winch control unit controls the winding and the unwinding of the suspension rope by the winch device. The load measurement device is connected to the suspension rope and measures a load by a suspended load that is suspended from the boom. The allowable deceleration rate derivation unit derives an allowable deceleration rate representing an allowable value of a deceleration rate of winding of the suspension rope, from a measured load. The measured load is a load by the suspended load, measured by the load measurement device. The allowable deceleration rate derivation unit derives the allowable deceleration rate that is decreased with a decrease in the measured load. The winch control unit decelerates the winding of the suspension rope by the winch device at a deceleration rate limited within a range equal to or less than the allowable deceleration rate.

(10) Also provided is a method for controlling a crane that includes the boom, the winch device, and the load measurement device. The method includes a deceleration allowance rate derivation step and a deceleration step. The deceleration allowance rate derivation step is a step of deriving an allowable deceleration rate from the measured load. The allowable deceleration rate represents an allowable value of a deceleration rate of the winding. In the deceleration allowance rate derivation

step, the allowable deceleration rate that is decreased with a decrease in the measured load is derived. The deceleration step is a step of decelerating the winding of the suspension rope by the winch device at a deceleration rate limited within a range equal to or less than the allowable deceleration rate.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a side view of a crane according to an embodiment of the present invention.
- (2) FIG. 2 is a block diagram representing elements for control in the crane.
- (3) FIG. 3 is a block diagram representing a configuration of a control device in the crane.
- (4) FIG. 4 is a flowchart showing an example of a winding deceleration control performed by the control device.
- (5) FIG. 5A is a graph showing an example of an allowable deceleration rate that is continuously decreased with a decrease in a suspension load.
- (6) FIG. 5B is a graph showing an example of an allowable deceleration rate that is decreased in multiple stages with a decrease in the suspension load.
- (7) FIG. 5C is a graph showing an example of an allowable deceleration rate that is varied in two stages with a decrease in the suspension load.
- (8) FIG. 6 is a graph showing a relationship between an allowable deceleration time according to the winding deceleration control and a first upper-limit winding speed.
- (9) FIG. 7 is a graph showing a relationship between a required stopping time according to the winding deceleration control and a second upper-limit winding speed.

### DETAILED DESCRIPTION

- (10) Below will be described an embodiment of the present invention with reference to the drawings. The following embodiment is illustrative of the invention, not intended to limit the scope of the invention.
- (11) FIG. 1 shows a crane **10** according to the embodiment of the present invention. The crane **10** is a work machine that hoists and moves a suspended load **9**. The crane **10** illustrated in FIG. 1 is a jib crane.
- (12) As shown in FIG. 1, the crane **10** includes a lower traveling body **11**, an upper slewing body **12**, a cab **13**, a gantry **15**, a winch device **16**, a counterweight **17**, a boom **21**, a derricking rope **31**, a suspension rope **32** and a hook **30**. The winch device **16** includes a first winch device **161** and a second winch device **162**.
- (13) The lower traveling body **11** is a pedestal part that supports the upper slewing body **12** slewably. The crane **10** is a mobile crane. Specifically, the lower traveling body **11** includes a traveling device **14**, which performs a traveling motion for moving the entire crane **10**. The traveling device **14** illustrated in FIG. 1 is a crawler type of traveling device. The lower traveling body **11** is an example of a lower base body.
- (14) The upper slewing body **12** is connected to the upper part of the lower traveling body **11** capably of slewing. The upper slewing body **12** is configured to support the cab **13**, the gantry **15**, and the winch device **16** so as to be slewed integrally with them. The gantry **15** is fixed to the upper slewing body **12** in a posture of projecting upward from the upper slewing body **12**. The upper slewing body **12** further supports the counterweight **17** and the boom **21**.
- (15) On the lower traveling body **11** is mounted a slewing motor **441** shown in FIG. 2, which drives the upper slewing body **12** to slew it (see FIG. 2). The cab **13** is an operation room. The boom **21** is connected to the upper slewing body **12** capably of derricking.
- (16) The derricking rope **31** is placed on a gantry sheave **23**, which is rotatably supported at the tip of the gantry **15**. The derricking rope **31** has opposite ends, which are connected to the distal end of

the boom **21** and the first winch device **161**, respectively.

(17) The first winch device **161** supports the boom **21** through the derricking rope **31**. The first winch device **161** is capable of winding and unwinding the derricking rope **31** to thereby change the derricking angle of the boom **21**. The first winch device **161** includes a first winch drum and a first winch motor **442** shown in FIG. 2. Around the first winch drum is wound the derricking rope **31**. The first winch motor **442** drives the first winch drum rotationally to thereby make the first winch drum wind and unwind the derricking rope **31**.

(18) The suspension rope **32** is placed on a point sheave **25**, which is rotatably supported at the distal end of the boom **21**. The hook **30** is connected to the tip of the suspension rope **32**.

(19) The suspended load **9** is engaged with the hook **30** to be thereby suspended from the distal end of the boom **21** through the suspension rope **32**. The suspended load **9** thus suspended exerts a downward suspension load **LD1** on the suspension rope **32**. The boom **21** supports the suspension rope **32** and the suspended load **9** against the suspension load **LD1** by the suspended load **9**.

(20) The second winch device **162** is capable of winding and unwinding the suspension rope **32** to thereby hoist and lower the hook **30** and the suspended load **9** engaged therewith. The second winch device **162** includes a second winch drum and a second winch motor **443** shown in FIG. 2. Around the second winch drum is wound the suspension rope **32**. The second winch motor **443** drives the second winch drum rotationally to thereby make the second winch drum wind and unwind the suspension rope **32**.

(21) The counterweight **17** is disposed so as to balance the weight of the counterweight **17** and the load by the boom **21**, the hook **30** and the suspended load **9** engaged therewith.

(22) The crane **10** includes a plurality of drive devices as shown in FIG. 2, an operation device **5**, a control device **6**, and a display device **7**, the plurality of driving devices including an engine **41**, a hydraulic pump **42**, a plurality of control valves **43**, and a plurality of hydraulic actuators **44**.

(23) The engine **41** is, for example, a diesel engine and drives the hydraulic pump **42**. The plurality of control valves **43** are interposed between the hydraulic pump **42** and the plurality of hydraulic actuators **44**, respectively, each configured to be opened and closed in response to a control signal that is input from the control device **6** to thereby render the flow of hydraulic fluid supplied from the hydraulic pump **42** to the plurality of hydraulic actuators **44** controllable.

(24) The plurality of hydraulic actuators **44** includes a plurality of hydraulic motors, which include the slewing motor **441**, the first winch motor **442**, and the second winch motor **443**.

(25) The operation device **5** and the display device **7** are devices for human interface, provided in the cab **13**. The crane **10** further includes a plurality of condition measurement devices **45** shown in FIG. 2, which measure respective conditions of a plurality of devices included in the crane **10**, respectively.

(26) The control device **6** is capable of communicating with a plurality of devices, which include the plurality of condition measurement devices **45** and the operation device **5**. The communication is performed through an in-vehicle network **100**, such as a Controller Area Network (CAN).

(27) The operation device **5** allows an operation to be applied to the operation device by an operator. The display device **7** is a device for displaying information, for example, a panel display device such as a liquid crystal display unit.

(28) The operation device **5** includes a slewing operation device **51**, a derricking operation device **52**, a lifting and lowering operation device **53**, and an information input device **54** shown in FIG. 2.

(29) The slewing operation device **51** includes a slewing lever **511** and a slewing signal output unit. The slewing lever **511** can be displaced in opposite directions from a neutral position by a slewing operation that is applied to the slewing lever **511** by an operator. The slewing signal output unit outputs a slewing instruction signal corresponding to the direction and magnitude (slewing operation amount) of the slewing operation applied to the slewing lever **511**. The slewing instruction signal is input to the control device **6** to instruct the rotational direction and the rotational speed of the slewing motor **441**.

(30) The derricking operation device **52** includes a derricking lever **521** and a derricking signal output unit. The derricking lever **521** can be displaced in opposite directions from a neutral position by a derricking operation that is applied to the derricking lever **521** by an operator. The derricking signal output unit outputs a derricking instruction signal corresponding to the direction and magnitude (derricking operation amount) of the derricking operation that is applied to the derricking lever **521**. The derricking instruction signal is input to the control device **6** to instruct a rotational direction and a rotational speed of the first winch motor **442**.

(31) The lifting and lowering operation device **53** includes a lifting and lowering lever **531** and a lifting and lowering signal output unit. The lifting and lowering lever **531** can be displaced in opposite directions from a neutral position by a lifting or lowering operation that is applied to the lifting and lowering lever **531** by an operator. The lifting and lowering signal output unit outputs a lifting or lowering signal corresponding to the direction and the magnitude (lifting operation amount) of the lifting or lowering operation that is applied to the lifting and lowering lever **531**. The lifting or lowering signal is input to the control device **6** to instruct the rotational direction and the rotational speed of the second winch motor **443**.

(32) The control device **6** inputs the control signal to the plurality of control valves **43** corresponding to the slewing motor **441**, the first winch motor **442** and the second winch motor **443**, respectively, in accordance with the slewing instruction signal, the derricking instruction signal, and the lifting or lowering instruction signal that are input from the slewing operation device **51**, the derricking operation device **52**, and the lifting and lowering operation device **53**, respectively.

(33) The information input device **54** allows information to be input to the information input device **54** by an operator. The information input device **54** may be, for example, a touch panel formed integrally with the display device **7**. The information input device **54** may, alternatively, be a device that allows information to be input to the information input device **54** through the voice of the operator.

(34) The plurality of condition measurement devices **45** include a load meter **451**, a derricking angle measurement device **452**, and an unwinding length measurement device **453** shown in FIG. 2. The result of respective measurements performed by the plurality of condition measurement devices **45** is transmitted to the control device **6** through the in-vehicle network **100**.

(35) The load meter **451** measures the load that is applied to the boom **21** by the suspended load **9**, namely, the suspension load **LD1** by the suspended load **9**. The load meter **451** is, for example, a load sensor such as a load cell attached to the derricking rope **31**. The load meter **451** is an example of a load measurement device.

(36) The derricking angle measurement device **452** measures the derricking angle of the boom **21**. The derricking angle measurement device **452** is, for example, an angle meter attached to the boom **21**.

(37) The unwinding length measurement device **453** is a device for measuring the unwinding length of the suspension rope **32**. The unwinding length is the length of the unwound portion of the suspension rope **32** from the second winch device **162**. The unwinding length measurement device **453** includes, for example, a rotor in contact with the suspension rope **32** to be rotated by following the movement of the suspension rope **32**, and a rotation detector that counts the number of rotations of the rotor to thereby determine the unwinding length of the suspension rope **32**.

(38) As shown in FIG. 3, the control device **6** includes a micro processing unit (MPU) **601**, a random access memory (RAM) **602**, a non-volatile memory **603** and a signal interface **604**. Each of the RAM **602** and the non-volatile memory **603** is a storage device that stores data readable by a computer.

(39) The MPU **601** is an example of a processor that executes a program stored in the non-volatile memory **603** to thereby carry out various data-processing and control.

(40) The RAM **602** is a volatile memory that temporarily stores the program to be executed by the

MPU **601** and the data to be derived or referenced by the MPU **601**.

(41) The non-volatile memory **603** previously stores the program to be executed by the MPU **601** and the data to be referenced by the MPU **601**. The non-volatile memory **603** is, for example, an electrically erasable programmable read only memory (EEPROM) or a flash memory.

(42) The signal interface **604** converts the measurement signal that is output from the condition measurement device **45** into digital data and transmits the digital data to the MPU **601**. The signal interface **604** further converts the control command that is output from the MPU **601** into a control signal such as a current signal or a voltage signal and inputs the signal to the device to be controlled.

(43) The MPU **601** of the control device **6** includes a plurality of processing modules that are realized by execution of a predetermined computer program. As shown in FIG. **2**, the plurality of processing modules include a main processing unit **61**, a slewing control unit **62**, a derricking control unit **63**, a lifting and lowering control unit **64**, and a suspension length derivation unit **65**.

(44) The main processing unit **61** executes a start control for starting various processing when the control device **6** is activated, a control of the display device **7**, and processing in accordance with the input of the information to the information input device **54**.

(45) The slewing control unit **62** inputs the control signal to the control valve **43** corresponding to the slewing motor **441**, among the plurality of control valves **43**, to thereby control the slewing direction and the slewing speed of the upper slewing body **12**.

(46) The derricking control unit **63** inputs the control signal to the control valve **43** corresponding to the first winch motor **442**, among the plurality of control valves **43**, to thereby control the unwinding and winding of the derricking rope **31** performed by the first winch device **161**. The derricking control unit **63**, thus, controls the derricking angle of the boom **21**.

(47) The lifting and lowering control unit **64** inputs a control signal to the control valve **43** corresponding to the second winch motor **443**, among the plurality of control valves **43**, to thereby control the unwinding and winding of the suspension rope **32** performed by the second winch device **162**. The lifting and lowering control unit **64**, thus, controls the height of the suspended load **9**.

(48) The lifting and lowering operation device **53** is an example of a winch operation device to which a winch operation for instructing the motion of the second winch device **162** is applied. The lifting and lowering control unit **64** is an example of a winch control unit that controls the winding and the unwinding by the second winch device **162** in accordance with the winch operation applied to the lifting and lowering operation device **53**.

(49) The suspension length derivation unit **65** derives a suspension length **L1** shown in FIG. **1** from the unwinding length measured by the unwinding length measurement device **453**, the predetermined length of the boom **21**, and the derricking angle measured by the derricking angle measurement device **452**. The suspension length **L1** is the length of the suspended portion of the suspension rope **32** from the distal end of the boom **21**.

(50) The unwinding length measurement device **453** and the suspension length derivation unit **65** are an example of suspension length measurement device that measures the suspension length **L1** of the suspension rope **32**. The lifting and lowering control unit **64** is capable of controlling the suspension length **L1** by means of inputting the control signal to the control valve **43** corresponding to the first winch motor **442** among the plurality of control valves **43**.

(51) The control device **6** performs a winding deceleration control. The winding deceleration control is a control of the deceleration of the winding of the suspension rope **32** performed by the second winch device **162**, for solving the following problems related to the winding. Rapid deceleration of the hoisting of the suspension rope **32** by the second winch device **162** with the suspended load **9** light may involve temporary looseness in the suspension rope **32** to cause an irregular winding in the second winch device **162**. Slow deceleration of the winding for preventing such irregular winding deteriorates the efficiency in work of carrying the suspended load **9** by the

crane **10**. The winding deceleration control prevents the irregular winding from being caused in the second winch device **162** by the above rapid deceleration of the winding of the suspension rope **32**, without significant deterioration in the work efficiency.

(52) The MPU **601** of the control device **6** further includes, as the processing module realized by the execution of the computer program, an allowable deceleration rate derivation unit **66**, a first upper-limit speed derivation unit **67**, and a second upper-limit speed derivation unit **68**, as shown in FIG. 2.

(53) The lifting and lowering control unit **64**, the allowable deceleration rate derivation unit **66**, the first upper-limit speed derivation unit **67**, and the second upper-limit speed derivation unit **68** execute the winding deceleration control.

(54) Below will be described an example of the winding deceleration control with reference to the flowchart shown in FIG. 4.

(55) The allowable deceleration rate derivation unit **66** starts the winding deceleration control, for example, when a measured load changes beyond a predetermined allowable range. The measured load is a load that is measured by the load meter **451**, namely, the suspension load **LD1**.

(56) In the winding deceleration control, the allowable deceleration rate derivation unit **66** acquires data of the measured load, i.e., the measured suspension load **LD1**, from the load meter **451**, and derives an allowable deceleration rate **dVL1** that is shown, for example, in any of FIGS. 5A, 5B and 5C, from the suspension load **LD1**. The allowable deceleration rate **dVL1** represents an allowable value of the deceleration rate of the winding of the suspension rope **32**.

(57) In the present embodiment, the allowable deceleration rate **dVL1** is a positive value. The larger the value of the allowable deceleration rate **dVL1**, therefore, the rapider deceleration of the winding of the suspension rope **32** is allowed. In other words, the smaller value of the allowable deceleration rate **dVL1** causes the winding of the suspension rope **32** to be required to be decelerated more slowly, that is, causes the deceleration rate to be limited more greatly.

(58) The allowable deceleration rate derivation unit **66** derives the allowable deceleration rate **dVL1** that is decreased with a decrease in the suspension load **LD1**. Each of FIGS. 5A, 5B, and 5C shows an example of a relationship between the suspension load **LD1** and the allowable deceleration rate **dVL1**.

(59) The allowable deceleration rate **dVL1** illustrated in FIG. 5A is decreased continuously with a decrease in the suspension load **LD1**. The allowable deceleration rate **dVL1** illustrated in FIG. 5B is decreased in multiple stages with a decrease in the suspension load **LD1**. The allowable deceleration rate **dVL1** illustrated in FIG. 5C is decreased in two stages with a decrease in the suspension load **LD1**.

(60) The allowable deceleration rate derivation unit **66** stores, for example, a calculation formula or a look-up table that specifies the relationship between the suspension load **LD1** and the allowable deceleration rate **dVL1** as described above, and applies the suspension load **LD1** thereto to derive the allowable deceleration rate **dVL1**.

(61) The control device **6** executes a step **S2** following the step **S1**. In the step **S2**, the first upper-limit speed derivation unit **67** derives a first upper-limit winding speed **Vmx1** from the allowable deceleration rate **dVL1** derived in the step **S1** and a predetermined allowable deceleration time **t1**.

(62) The first upper-limit speed derivation unit **67** calculates a first starting speed **Vs1** shown in FIG. 6 as the first upper-limit winding speed **Vx1**. The first starting speed **Vs1** is such a speed of the winding at the starting of deceleration that the allowable deceleration time **t1** is required for the stop of the winding of the suspension rope **32** by the second winch device **162** when the winding of the suspension rope **32** by the second winch device **162** is decelerated at the allowable deceleration rate **dVL1** from the first starting speed **Vs1**.

(63) For example, the first upper-limit speed derivation unit **67** derives the first upper-limit winding speed **Vmx1** based on the following equation (1):

$$t1=Vmx1/dVL1 \quad (1)$$



(64) The control device **6** executes a step **S3** following the step **S2**. In the step **S3**, the lifting and lowering control unit **64** judges whether or not the winding of the suspension rope **32** by the second winch device **162** is being performed. Incidentally, the lifting and lowering control unit **64** executes the control of the second winch device **162** in accordance with the lifting or lowering operation applied to the lifting and lowering operation device **53** in parallel with the processing in and after the step **S3**.

(65) When judging that the winding is being performed (YES in the step **S3**), the lifting and lowering control unit **64** executes the processing in and after a step **S4**. This processing is to control the deceleration of the winding of the second winch device **162** in accordance with the lifting or lowering operation applied to the lifting and lowering operation device **53** within a range of the winding speed not exceeding the first upper-limit winding speed  $V_{mx1}$  or a second upper-limit winding speed  $V_{mx2}$  shown in FIG. 7.

(66) The lifting and lowering control unit **64** derives the time change rate of the unwinding length measured by the unwinding length measurement device **453** and the time change rate of the time change rate thereof, thereby determining the speed and the acceleration of the winding of the suspension rope **32**.

(67) The second upper-limit speed derivation unit **68** derives the second upper-limit winding speed  $V_{mx2}$  shown in FIG. 7 from the allowable deceleration rate  $dVL1$  derived in the step **S1**, the suspension length  $L1$  derived by the suspension length derivation unit and a minimum suspension length  $L0$ , in the step **S4**. The minimum suspension length  $L0$  is the minimum value of the suspension length  $L1$ , being preset by the main processing unit **61**, for example, based on information that is input to the information input device **54**. The suspension length  $L1$  is measured by the suspension length measurement device constituted by the unwinding length measurement device **453** and the suspension length derivation unit **65**.

(68) The second upper-limit speed derivation unit **68** derives a second starting speed  $Vs2$  shown in FIG. 7 as the second upper-limit winding speed  $V_{mx2}$ . The second starting speed  $Vs2$  is such a speed of the winding at the start of deceleration that the suspension length  $L1$  is decreased from the current measuring length to the minimum suspension length  $L0$  in the period until the stop of the winding by the second winch device **162**, when the winding of the suspension rope **32** by the second winch device **162** is decelerated from the second starting speed  $Vs2$  at the allowable deceleration rate  $dVL1$ .

(69) FIG. 7 shows the required stopping time  $t2$  corresponding to the second upper-limit winding speed  $Vx2$  and a winding length  $LUP1$ . The required stopping time  $t2$  is a time required for the stop of the winding of the suspension rope **32** by the second winch device **162** when the winding is decelerated from the second starting speed  $Vs2$  at the allowable deceleration rate  $dVL1$ . The winding length  $LUP1$  is the length of the wound portion of the suspension rope **32** that is wound during the decrease in the suspension length  $L1$  from the current measurement length to the minimum suspension length  $L0$ .

(70) For example, the second upper-limit speed derivation unit **68** derives the second upper-limit winding speed  $V_{mx2}$  based on the following equation (2):

$$V_{mx2} = (2 \times dVL1) \cdot \text{sup.} 0.5 \quad (2).$$

(71) The control device **6** executes a step **S5** following the step **S4**. In the step **S5**, the lifting and lowering control unit **64** executes a winding speed limit control. The winding speed limit control is a control for limiting the speed of the winding of the suspension rope **32** by the second winch device **162** within a range equal to or less than the first upper-limit winding speed  $V_{mx1}$  and within a range equal to or less than the second upper-limit winding speed  $V_{mx2}$ . In this control, when the winding speed of the suspension rope **32** corresponding to the lifting or lowering operation applied to the lifting and lowering operation device **53** is equal to or less than the first upper-limit winding speed  $V_{mx1}$  and equal to or less than the second upper-limit winding speed  $V_{mx2}$ , the lifting and lowering control unit **64** controls the speed of the winding by the second winch device **162** in

accordance with the lifting or lowering operation. When the winding speed of the suspension rope **32** corresponding to the lifting or lowering operation applied to the lifting and lowering operation device **53**, conversely, exceeds at least one of the first upper-limit winding speed  $V_{mx1}$  and the second upper-limit winding speed  $V_{mx2}$ , the lifting and lowering control unit **64** controls the winding by the second winch device **162** so as to render the winding speed of the suspension rope **32** equal to the lower speed selected from the first upper-limit winding speed  $V_{mx1}$  and the second upper-limit winding speed  $V_{mx2}$ .

(72) The control device **6** executes a step **S6** following the step **S5**. In the step **S6**, the lifting and lowering control unit **64** judges whether or not a deceleration operation for decelerating the winding of the suspension rope **32** is applied to the lifting and lowering operation device **53**. Only when judging that the deceleration operation is applied (YES in the step **S6**), the lifting and lowering control unit **64** executes the deceleration rate limit control of the step **S7**.

(73) The deceleration rate limit control is a control for limiting the deceleration rate of the deceleration of the winding of the suspension rope **32** by the second winch device **162** within a range equal to or less than the allowable deceleration rate  $dVL1$ . The lifting and lowering control unit **64** can limit the deceleration rate of the winding of the suspension rope **32** within a range equal to or less than the allowable deceleration rate  $dVL1$ , for example, by means of inputting a control signal for feedback control to the control valve **43** corresponding to the second winch motor **443** among the plurality of control valves **43** on the basis of the acceleration of the winding of the suspension rope **32**.

(74) The lifting and lowering control unit **64**, in step **S8**, judges whether or not the winding by the second winch device **162** has been stopped, regardless of the performance of the deceleration rate limit control, and continues the winding deceleration control (the speed limit control or both of the speed limit control and the deceleration rate limit control) until the winding is judged to be stopped (NO in the step **S8**). Upon judging that the winding by the second winch device **162** has been stopped (YES in the step **S8**), the lifting and lowering control unit **64** terminates the winding deceleration control.

(75) The lifting and lowering control unit **64** and the allowable deceleration rate derivation unit **66**, thus, can prevent the irregular winding from being caused in the second winch device **162** by the rapid deceleration of the winding of the suspension rope **32**, by means of limiting the deceleration of the winding of the suspension rope **32** more greatly with a decrease in the suspension load  $LD1$ , as shown in FIGS. 5A to 5C. On the other hand, the lifting and lowering control unit **64** and the allowable deceleration rate derivation unit **66** can restrain the efficiency in the work of carrying the suspended load **9** by the crane **10** from being unnecessarily deteriorated, by means of reducing or releasing the limitation of the deceleration of the winding by the suspension rope **32** when the suspension load  $LD1$  is large, that is, when the irregular winding is less likely to occur.

(76) Moreover, limiting the winding speed of the suspension rope **32** to the first upper-limit winding speed  $V_{mx1}$  or less, the lifting and lowering control unit **64** according to the embodiment enable the required stopping time required for the stop of the winding of the suspension rope **32** after the start of the deceleration to be confined within a predetermined allowable deceleration time  $t1$ . This prevents the required stopping time from being excessively prolonged by the deceleration limit control (step **S7**).

(77) Furthermore, limiting the winding speed of the suspension rope **32** to the second upper-limit winding speed  $V_{mx2}$  or less, the lifting and lowering control unit **64** according to the embodiment prevents the suspended load **9** from being hoisted beyond the height corresponding to the minimum suspension length  $L0$ , regardless of the limitation of the deceleration rate. The control based on the second upper-limit winding speed  $V_{mx2}$  is useful for the case of requiring the restriction of the lifting height of the suspended load **9**, for example, prevention of the suspended load **9** from being raised to the vicinity of the distal end of the boom **21**.

(78) The crane according to the present invention is not limited to the above-described

embodiments. For example, the crane **10** is modifiable as follows.

(79) In the crane **10**, it is also possible that: the information input device **54** is configured to allow a mode selection operation to be input to the information input device **54**; the main processing unit **61** is configured to select an action mode corresponding to the mode selection operation from among a plurality of preset operation modes for the speed limit control; and the lifting and lowering control unit **64** is configured to determine execution or non-execution of the speed limit control and determine contents of the speed limit control, according to the selected operation mode. The plurality of action modes include, for example, a mode of omitting one or both of the winding speed limit control based on the first upper-limit winding speed  $V_{x1}$  and the winding speed limit control based on the second upper-limit winding speed  $V_{x2}$ , and a mode of performing both the controls.

(80) In the crane **10**, processing modules and controls for one or both of the first upper-limit winding speed  $V_{x1}$  and the second upper-limit winding speed  $V_{x2}$  are omissible.

(81) The allowable deceleration rate derivation unit **66** may derive a value corresponding to the allowable deceleration rate  $dVL1$ , which is not limited to the value of the allowable deceleration rate  $dVL1$  itself. The allowable deceleration rate derivation unit **66**, for example, may be configured to derive a winding target speed that gradually changes from the current speed to the stop of the winding of the suspension rope **32**, and the lifting control unit **64** may be configured to perform a control to render the actual winding speed closer to the winding target speed. The winding target speed is the target speed that is set in consideration with the allowable deceleration rate  $dVL1$ .

(82) Thus is provided a crane and crane control method that are capable of preventing an irregular winding from being caused in a winch device by rapid deceleration of the winding of a suspension rope without a significant deterioration in work efficiency.

(83) Provided is a crane including a boom, a winch device, a winch control unit, a load measurement device, and an allowable deceleration rate derivation unit. The boom supports a suspension rope suspended from the boom. The winch device is configured to perform winding and unwinding of the suspension rope. The winch control unit controls the winding and the unwinding of the suspension rope by the winch device. The load measurement device is connected to the suspension rope and measures a load by a suspended load that is suspended from the boom. The allowable deceleration rate derivation unit derives, from a measured load, an allowable deceleration rate representing an allowable value of a deceleration rate of winding of the suspension rope. The measured load is a load by the suspended load, measured by the load measurement device. The allowable deceleration rate derivation unit derives the allowable deceleration rate that is decreased with a decrease in the measured load. The winch control unit decelerates the winding of the suspension rope by the winch device at a deceleration rate limited within a range equal to or less than the allowable deceleration rate.

(84) Also provided is a method for controlling a crane that includes the boom, the winch device, and the load measurement device. The method includes a deceleration allowance rate derivation step and a deceleration step. The deceleration allowance rate derivation step is a step of deriving an allowable deceleration rate from the measured load. The allowable deceleration rate represents an allowable value of a deceleration rate of the winding. In the deceleration allowance rate derivation step, the allowable deceleration rate that is decreased with a decrease in the measured load is derived. The deceleration step is a step of decelerating the winding of the suspension rope by the winch device at a deceleration rate limited within a range equal to or less than the allowable deceleration rate.

(85) According to the crane and the control method, a small allowable deceleration rate is derived from the measured load, when the measured load is small, to greatly limit the deceleration rate of the winding of the suspension rope, which prevents irregular winding from being caused by the rapid deceleration of the winding; on the other hand, a large allowable deceleration rate is derived

from the measured load, when the measured load is large, to reduce or release the limitation of the deceleration rate, which restrains work efficiency from being unnecessarily deteriorated.

(86) It is preferable that the crane further includes a first upper-limit winding speed derivation unit that derives a first upper-limit winding speed from the allowable deceleration rate and a predetermined allowable deceleration time and the winch control unit is configured to limit the speed of the winding of the suspension rope by the winch device within a range equal to or less than the first upper-limit winding speed. The first upper-limit winding speed is a speed that renders a time required for the stop of the winch device after the start of the deceleration of the winding of the suspension rope by the winch device at the allowable deceleration rate, when the speed of winding is the first upper-limit winding speed, equal to the allowable deceleration time. The winding speed limit control based on the first upper-limit winding speed can prevent the time required for the stop of the winding by the winch device from being excessively prolonged by the limitation of the deceleration rate.

(87) Preferably, the crane further includes a suspension length measurement device that measures a suspension length that is a length of a portion of the suspension rope, which portion is suspended from the boom, and a second upper-limit winding speed derivation unit that derives a second upper-limit winding speed from the allowable deceleration rate, a measured suspension length that is the suspension length measured by the suspension length measurement device, and a preset minimum suspension length, wherein the winch control unit is configured to limit the speed of the winding of the suspension rope by the winch device within a range equal to or less than the second upper-limit winding speed. The second upper-limit winding speed is a speed that causes the suspension length to be decreased from the measured suspension length to the minimum suspension length in a period from start of the deceleration of the winding at the allowable deceleration rate, when the speed of winding of the suspension rope by the winch device is the second upper-limit winding speed, to the stop of the winch device. The winding speed limit control based on the second upper-limit winding speed can prevent the suspension length from being excessively small upon the stop of the winding by the winch device.

## Claims

1. A crane, comprising: a boom that supports a suspension rope suspended from the boom; a winch device configured to perform winding and unwinding of the suspension rope; a winch control unit that controls the winding and the unwinding of the suspension rope by the winch device; a load measurement device that is connected to the suspension rope and measures a load by a suspended load that is suspended from the boom; and an allowable deceleration rate derivation unit that derives an allowable deceleration rate that represents an allowable value of a deceleration rate of the winding of the suspension rope, from a measured load that is a load by the suspended load and measured by the load measurement device, wherein: the allowable deceleration rate derivation unit derives the allowable deceleration rate that is decreased with a decrease in the measured load; and the winch control unit decelerates the winding of the suspension rope by the winch device at a deceleration rate limited within a range equal to or less than the allowable deceleration rate.
2. The crane according to claim 1, further comprising a first upper-limit winding speed derivation unit that derives a first upper-limit winding speed from the allowable deceleration rate and a predetermined allowable deceleration time, the first upper-limit winding speed being a speed that renders a time required for stop of the winch device after start of the deceleration of the winding of the suspension rope by the winch device at the allowable deceleration rate, when the speed of winding is the first upper-limit winding speed, equal to the allowable deceleration time, wherein the winch control unit is configured to limit the speed of the winding of the suspension rope by the winch device within a range equal to or less than the first upper-limit winding speed.
3. The crane according to claim 1, further comprising: a suspension length measurement device that

measures a suspension length that is a length of a portion of the suspension rope, the portion being suspended from the boom, and a second upper-limit winding speed derivation unit that derives a second upper-limit winding speed from the allowable deceleration rate, a measured suspension length that is the suspension length measured by the suspension length measurement device, and a preset minimum suspension length, the second upper-limit winding speed being a speed that causes the suspension length to be decreased from the measured suspension length to the minimum suspension length in a period from start of the deceleration of the winding at the allowable deceleration rate, when the speed of winding of the suspension rope by the winch device is the second upper-limit winding speed, to the stop of the winch device, wherein the winch control unit is configured to limit the speed of the winding of the suspension rope by the winch device within a range equal to or less than the second upper-limit winding speed.

4. A method for controlling a crane that includes a boom that supports a suspension rope suspended from the boom, a winch device configured to wind and unwind the suspension rope, and a load measurement device that is connected to the suspension rope and measures a load by a suspended load suspended from the boom, the method comprising: an allowable deceleration rate derivation step of deriving an allowable deceleration rate from a measured load that is a suspension load measured by the load measurement device, the allowable deceleration rate representing an allowable value of a deceleration rate of the winding and decreased with a decrease in the measured load; and a deceleration step of decelerating the winding of the suspension rope by the winch device at a deceleration rate limited within a range equal to or less than the allowable deceleration rate.

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